Chapter 2 Note Packet (Key) (4)

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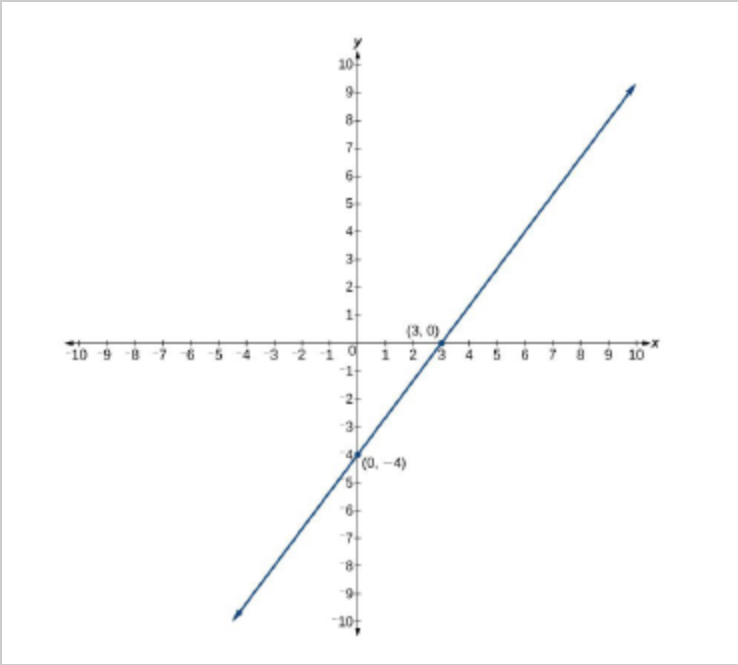


▶ *View Description*

8-6-OT-

10

+6

8+

7

(3,0)

1 2 3 4 5 6 7 8 9 10

6+

5

4+

3+

2+

1+

7

6-5

3210

-1+

-2+

-3+

(0,-4)

st

6+

7

+6

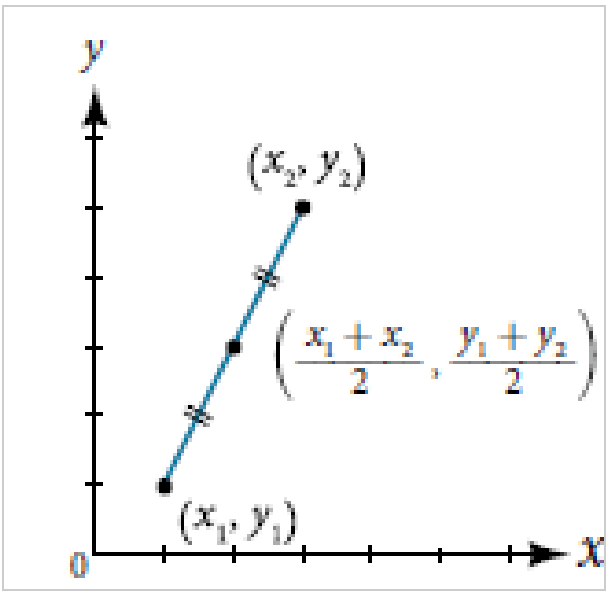
8.

-10+

Midpoint Formula

The midpoint of the line segment from (x_1, y_1) to (x_2, y_2) is

$$\left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right)$$



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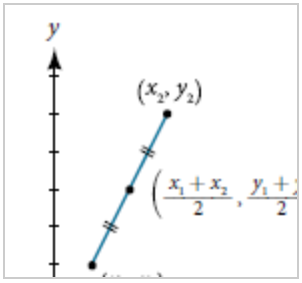

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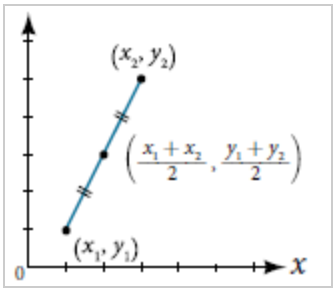

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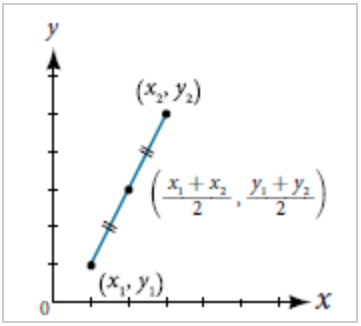

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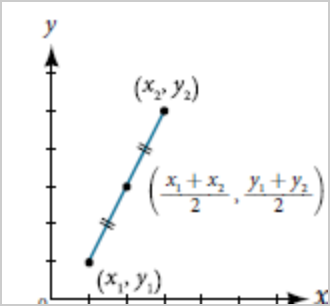

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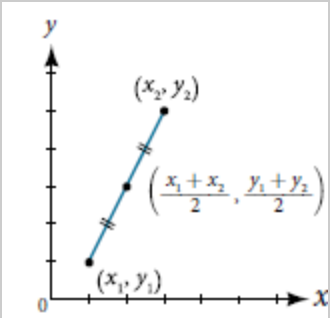

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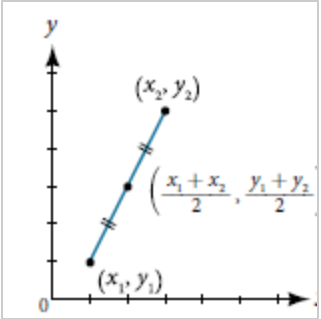

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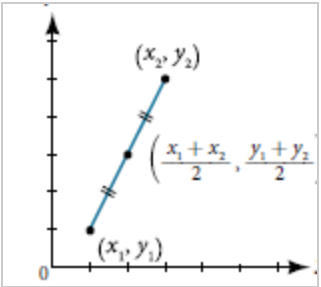

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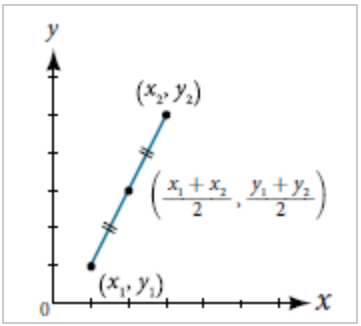

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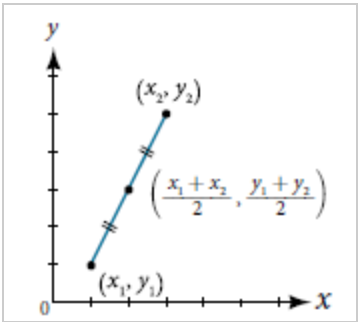

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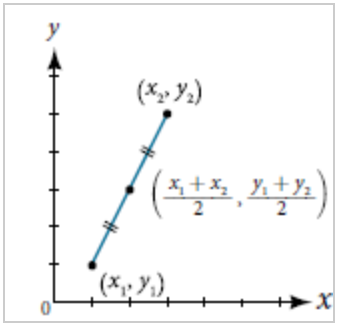

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10. $\frac{1}{2} - \frac{1}{3}x = \frac{4}{3}$

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$\frac{1}{2} - \frac{1}{3}x = \frac{4}{3}$ Rewrite with a

$\frac{3}{3} \cdot \frac{1}{2} - \frac{2}{2} \cdot \frac{x}{3} = \frac{2}{2} \cdot \frac{4}{3}$ Multiply

$\frac{1}{2} - \frac{1}{3}x = \frac{4}{3}$ Rewrite with a common denominator

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$\frac{3}{6} - \frac{2x}{6} = \frac{8}{6}$ Multiply both sides by 6

$\frac{3}{6} - \frac{2x}{6} = \frac{8}{6}$ Multipl

$3 - 2x = 8$ Subtract

denominator

$-2x = 5$ Divide botl

$x = \frac{-5}{2}$

$3 - 2x = 8$ Subtract 3 from both sides

$\frac{3}{6} - \frac{2x}{6} = \frac{8}{6}$ Multiply b

$3 - 2x = 8$ Subtract 3 fr

denominator

$-2x = 5$ Divide both si

$x = \frac{-5}{2}$

$-2x = 5$ Divide both sides by -2

denominator

$$\frac{3}{6}-\frac{2x}{6}=\frac{8}{6} \quad \text{Multipl}$$
$$3-2x=8 \quad \text{Subtract}$$
$$-2x=5 \quad \text{Divide both}$$
$$x=\frac{-5}{2}$$

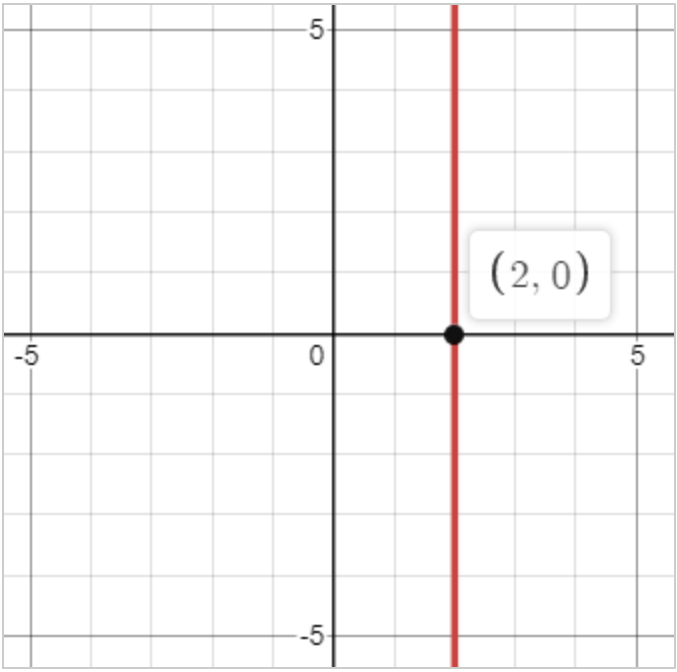
$x = \frac{-5}{2}$

denominator

$$\frac{3}{6}-\frac{2x}{6}=\frac{8}{6} \quad \text{Multiply}$$
$$3-2x=8 \quad \text{Subtract 3}$$
$$-2x=5 \quad \text{Divide both}$$
$$x=\frac{-5}{2}$$

Equations of Lines

1. $x = 2$



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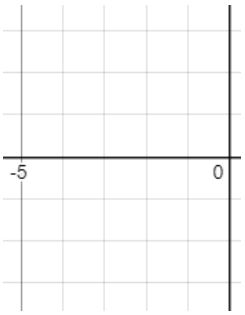
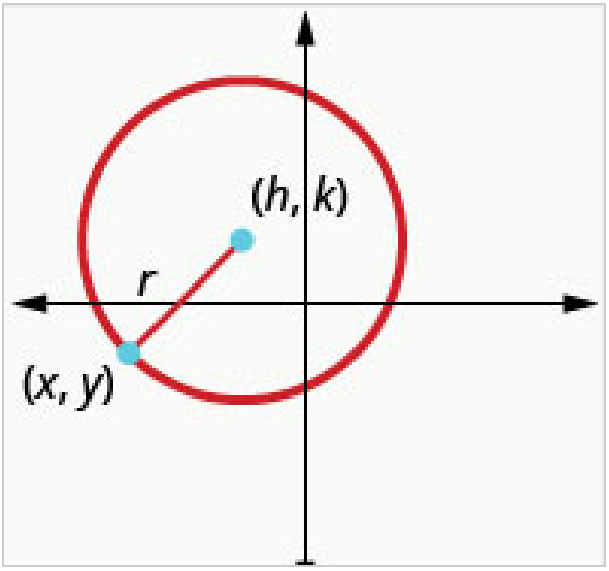
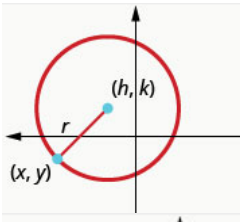
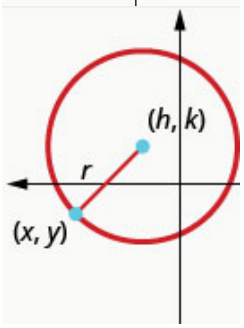
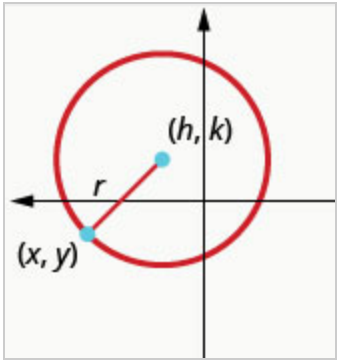


Equation of a Circle









▶ *View Description*

The **standard form of the equation of a circle** with center (h, k) and radius r is

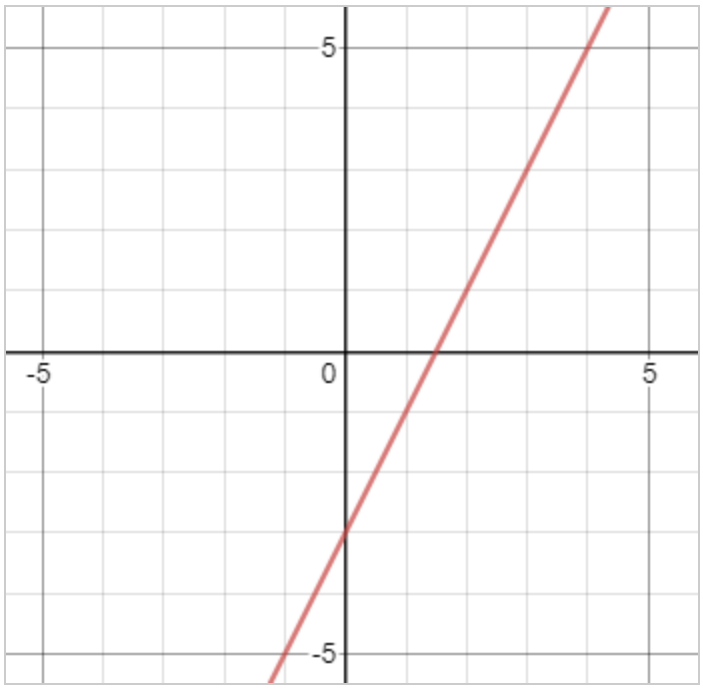
$$(x-h)^2 + (y-k)^2 = r^2$$

The **general form of the equation of a circle** is

$$Ax^2 + Ay^2 + Dx + Ey + F = 0,$$

where $A \neq 0$.

Graph the linear function



▶ *View Description*

$$\sqrt{25(-1)} = \frac{5}{\sqrt{25}}\frac{i}{\sqrt{-1}} = 5i$$

$$\sqrt{25(-1)} = \overbrace{\sqrt{25}}^{5}$$

$$\sqrt{25(-1)} = \overbrace{\sqrt{25}}^{5}\overbrace{\sqrt{-1}}^{i} =$$

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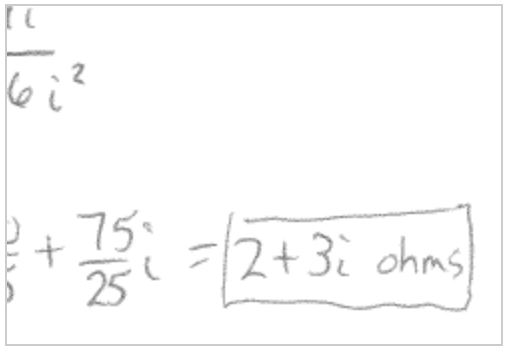
$$Z = \frac{V}{I} = \frac{(18+i)(3+4i)}{(3-4i)(3+4i)} = \frac{54+72i+3i+4i^2}{9+12i-12i-16i^2}$$

$$= \frac{54+72i+3i+4}{9+\cancel{12i}-\cancel{12i}-16}$$

$$= \frac{50+75i}{50}$$

▶ View Description

$$= \frac{54+75i+4(-1)}{9-16(-1)} = \frac{50+75i}{25} = \frac{50}{25} + \frac{75}{25}i =$$

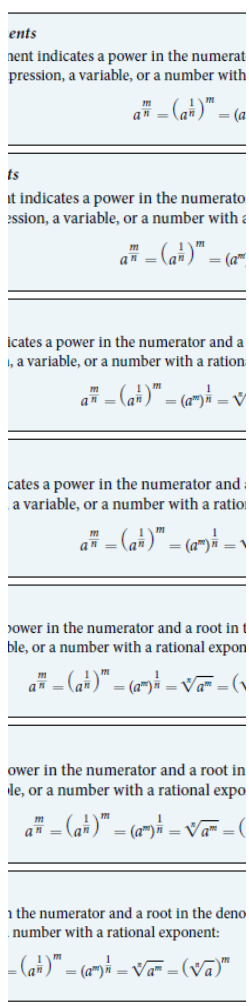


▶ View Description

$[2 + 3i \text{ ohms}]$

rational exponents

A rational exponent indicates a power in the numerator and a root in the denominator. There are multiple ways of writing an expression, a variable, or a number with a rational exponent:



the numerator and a root in the den
number with a rational exponent:

$= \left(a^{\frac{1}{n}}\right)^m = (a^m)^{\frac{1}{n}} = \sqrt[n]{a^m} = \left(\sqrt[n]{a}\right)^m$

merator and a root in the denominator
with a rational exponent:

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nerator and a root in the denominat
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$= (a^m)^{\frac{1}{n}} = \sqrt[n]{a^m} = \left(\sqrt[n]{a}\right)^m$

$$a^{\frac{m}{n}} = (a^{\frac{1}{n}})^m = (a^m)^{\frac{1}{n}} = \sqrt[n]{a^m} = (\sqrt[n]{a})^m$$

radical equations

An equation containing terms with a variable in the radicand is called a radical equation.

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▶ *View Description*

the distance formula Given endpoints (x_1, y_1) and (x_2, y_2), the distance between two points is given by

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

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$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$

▶ *View Description*

16. $\frac{3}{x} - \frac{1}{3} = \frac{1}{6}$

$\frac{3}{x} - \frac{1}{3} = \frac{1}{6}$ Multiply both sides by 6x

$6x \cdot \frac{3}{x} - 6x \cdot \frac{1}{3} = 6x \cdot \frac{1}{6}$ Simplify

$18 - 2x = x$ Add 2x to both sides

$18 = 3x$ Divide both sides by 3

$x \neq 0; x = 6$

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$18 = 3x$ Divide both sides by 3

$x \neq 0;\ x = 6$

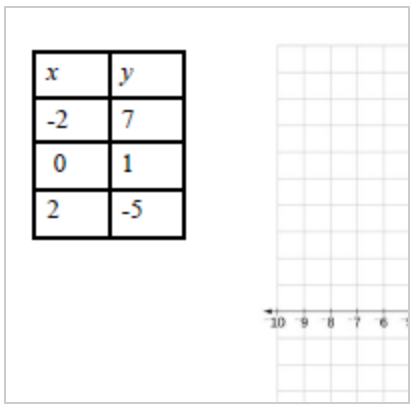
=4+(-3)i Real: 4
Imag.: -3

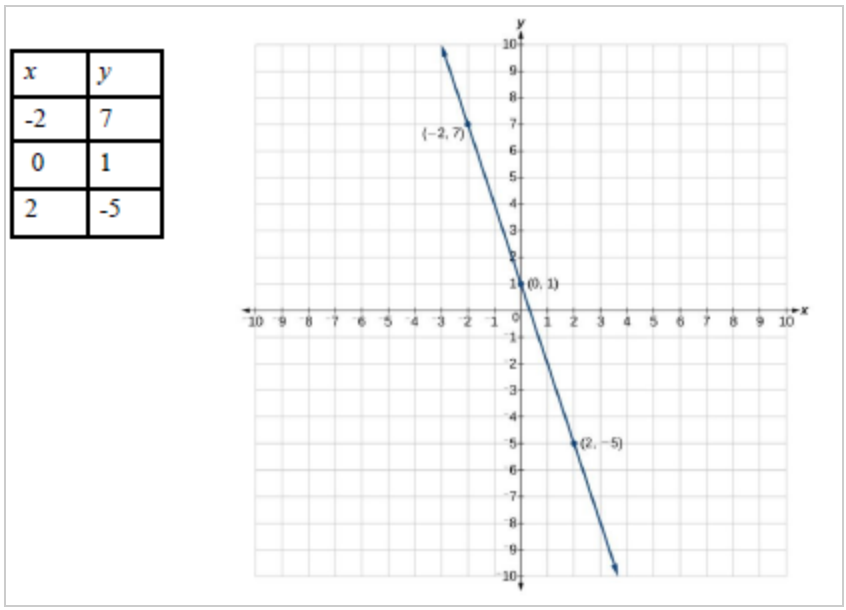
$\frac{4}{5}+\frac{7}{5}i$ Real: $\frac{4}{5}$
Imag: $\frac{7}{5}$

$\frac{3}{8}+0i$ Real: $\frac{3}{8}$
Imag: 0

quadratic form If the exponent on the middle term is one-half of the exponent on the leading term, we have an equation in **quadratic form**, which we can solve as if it were a quadratic. We substitute a variable for the middle term to solve equations in quadratic form.
Data Table

x	y
-2	7
0	1
2	-5



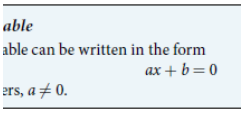


▶ View Description

linear equation in one variable A linear equation in one variable can be written in the form

$$ax + b = 0$$

where a and b are real numbers, a



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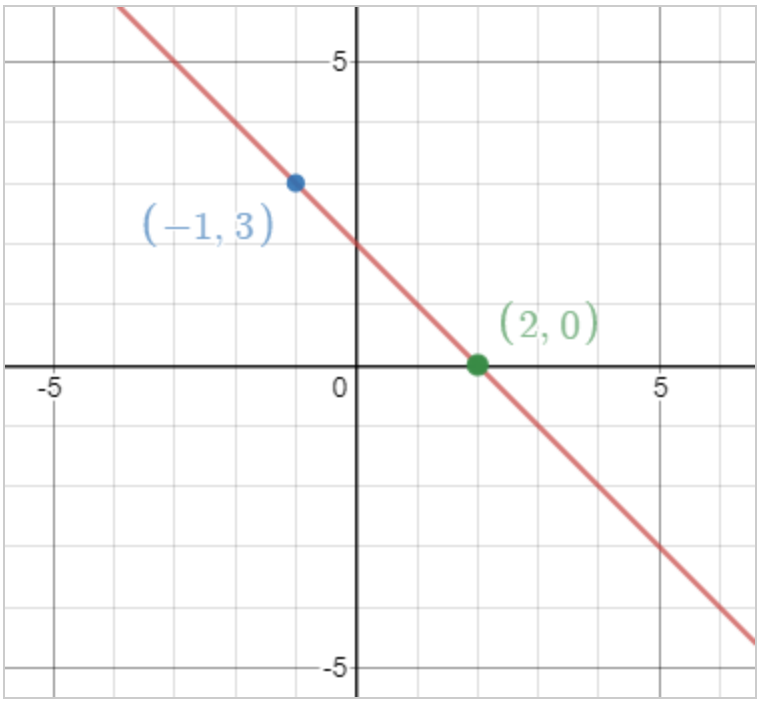

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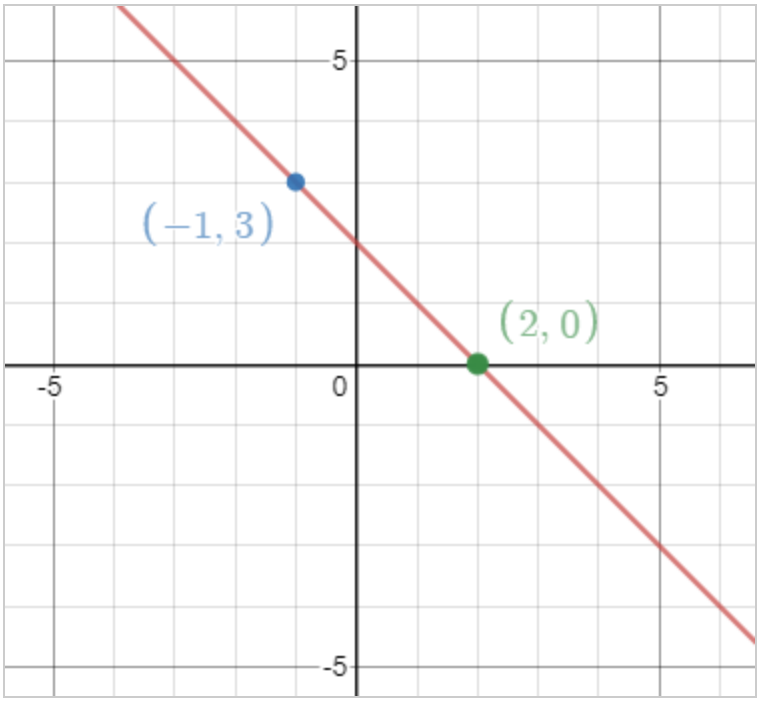

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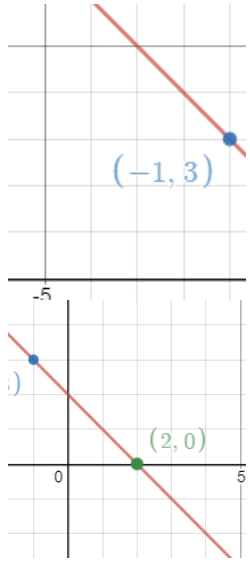

35. (-1,3) and (4,-5)

$m = \frac{-5-3}{4-(-1)}$

35. $(-1,3)$ and $(4,-5)$

$m = \frac{-5-3}{4-(-1)}$ Subtract

$m = \frac{-8}{5}$ Use the slope and

$y-3 = -\frac{8}{5}(x-(-1))$ Rewri

Subtract

$m = \frac{-8}{5}$

35. $(-1,3)$ and $(4,-5)$

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Use the slope and a point in the point-slope formula

$y - 3 = -\frac{8}{5}(x - (-1))$

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$m = \frac{-8}{5}$ Use the slope and

$y-3 = -\frac{8}{5}(x-(-1))$ Rewri

Rewrite

$y - 3 = -\frac{8}{5}(x + 1)$ Distribute the -8/5

$y - 3 = -\frac{8}{5}x - \frac{8}{5}$

he point-slope formula

$y - 3 = -\frac{8}{5}(x$

$y - 3 = -\frac{8}{5}x -$

$y = -\frac{8}{5}x + \frac{7}{5}$

he point-slope formula

$y - 3 = -\frac{8}{5}(x$

$y - 3 = -\frac{8}{5}x -$

$y = -\frac{8}{5}x + \frac{7}{5}$

Add 3 to both sides

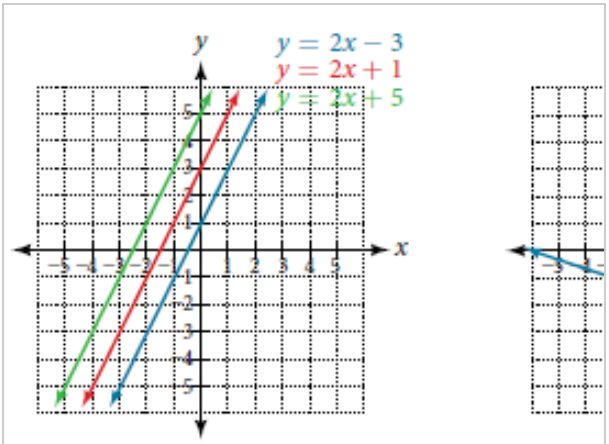
$y = -\frac{8}{5}x + \frac{7}{5}$

he point-slope formula

$y - 3 = -\frac{8}{5}(x$

$y - 3 = -\frac{8}{5}x -$

$y = -\frac{8}{5}x + \frac{7}{5}$



Parallel lines

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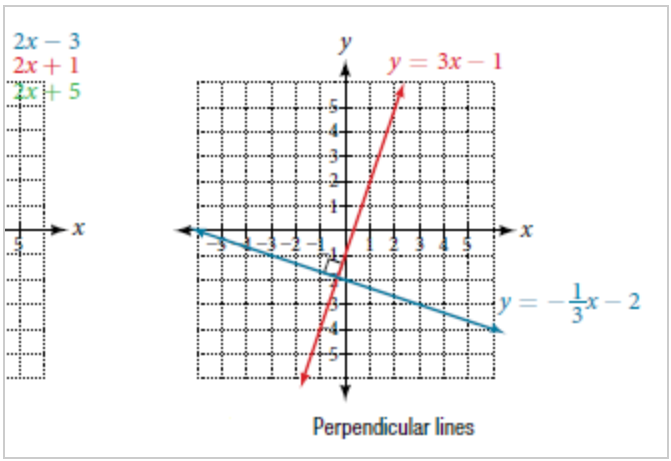
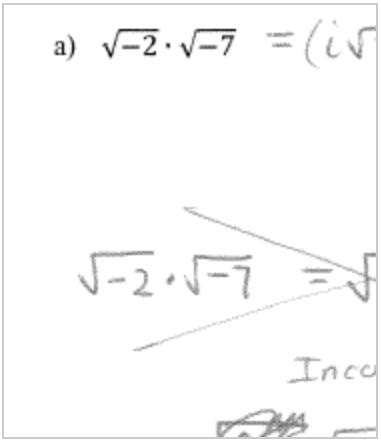


▶ *View Description*

y $y = 2x - 3$ $y = 2x + 1$ $y = 2x + 5$ x Parallel lines y $y = 3x - 1$ $y = -\frac{1}{3}x - 2$ x Perpendicular lines a) $\sqrt{-2} \cdot \sqrt{-7} = (i\sqrt{2})(i\sqrt{7}) = i^2\sqrt{2}\sqrt{7} = -\sqrt{14}$

$\boxed{-\sqrt{14}}$

i^2 -1 $\sqrt{2 \cdot 7}$



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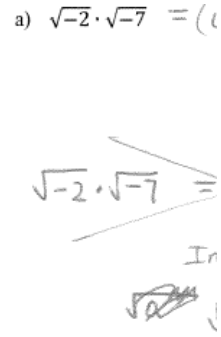
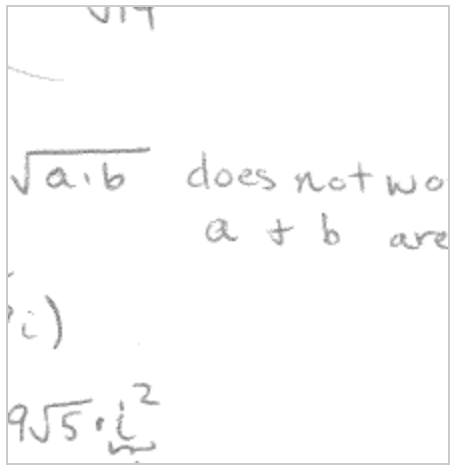
$\sqrt{-2} \cdot \sqrt{-7} = \sqrt{(-2)(-7)} = \sqrt{14}$ Incorrect

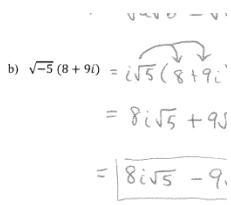


$\sqrt{a}\sqrt{b} = \sqrt{a \cdot b}$ does not work when $a + b$ are both negative. b) $\sqrt{-5}(8 + 9i) = i\sqrt{5}(8 + 9i)$

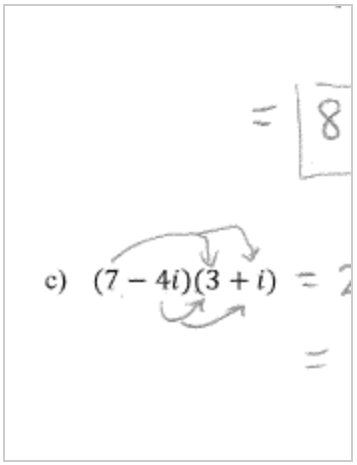


▶ View Description

$= 8i\sqrt{5} + 9\sqrt{5} \cdot i^2 \; -1 = 8i\sqrt{5} - 9\sqrt{5}$

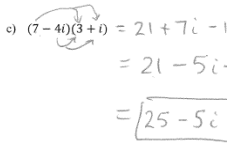


c) $(7 - 4i)(3 + i) = 21 + 7i - 12i - 4i^2$



▶ View Description

$= 21 - 5i - 4(-1) = 25 - 5i$



$a + bi$

polynomial equations

A polynomial of degree *n* is an expression of the type

lynomial equations

olynomial of degree n is an expression of the type

$$a_n x^n + a_{n-1} x^{n-1} + \cdots + a_2 x^2 + a_1 x + a_0$$

ere n is a positive integer and $a_n, \ldots, a_0$ are real numbers and $a_n \neq 0$.

tting the polynomial equal to zero gives a **polynomial equation**. The total number of solutions (real ar mplex) to a polynomial equation is equal to the highest exponent n.

▶ *View Description*

$$a_n x^n + a_{n-1} x^{n-1} + \cdots + a_2 x^2 + a_1 x + a_0$$

where *n* is a positive Integer and $a_n, \ldots, a_0$ are real numbers and

ion of the type

$x^n + a_{n-1} x^{n-1} + \cdots + a_2 x^2 + a_1 x + a_0$

$, a_0$ are real numbers and $a_n \neq 0$.

gives a **polynomial equation**. The total numb

equal to the highest exponent n.

$a_n \neq 0$. Setting the polynomial equal to zero gives a **polynomial equation**. The total number of solutions (real and complex) to a polynomial equation is equal to the highest exponent *n*.

$\sqrt{x-1} = x - 7$

$$\sqrt{x-1} = x-7$$
$$\left(\sqrt{x-1}\right)^2 = (x-7)^2$$
$$x-1 = (x-7)(x-7)$$
$$x-1 = x^2 - 7x - 7x + 49$$
$$x-1 = x^2 - 14x + 49$$

Square both sides

$(\sqrt{x-1})^2 = (x-7)^2$

$$\sqrt{x-1} = x-7$$
$$\left(\sqrt{x-1}\right)^2 = (x-7)^2$$
$$x-1 = (x-7)(x-7)$$
$$x-1 = x^2 - 7x - 7x + 49$$
$$x-1 = x^2 - 14x + 49$$
$$x^2 - 15x + 50 = 0$$
$$(x^2 - 10x) + (-5x + 50) =$$
$$x(x-10) - 5(x-10) = 0$$
$$(x-5)(x-10) = 0$$

Expand

x-1=(x-7)(x-7) Use the FOIL method

x-1=x^2-7x-7x+49 Combine like terms

x-1=x^2-14x+49 Rewrite

x^2-15x+50=0 Rewrite -15x as -10x - 5x

(x^2-10x)+(-5x+50)=0 Factor out the GCF of each expression

x(x-10)-5(x-10) = 0 Use the distributive property

(x-5)(x-10)=0 Use the zero-product property

x-5=0 Solve for x

x=5

x-10=0

x = 10

$\sqrt{5-1} = 5-7$

$(x-5)(x-10)=0$

$x-5=0$

$x=5$

$x-10=0$

$x=10$

$\sqrt{5-1}=5-7$

$\sqrt{4}=-2$

$2 \neq -2$

$\sqrt{10-1}=10-7$

$\sqrt{9}=3$

$3=3$

$x-5=0$

$x=5$

$x-10=0$

$x=10$

$\sqrt{5-1}=5-7$

$\sqrt{4}=-2$

$2 \neq -2$

$\sqrt{10-1}=10-7$

$\sqrt{9}=3$

$3=3$

$x=10$

Substitute to check answer

$\sqrt{4} = -2$

2≠-2 5 is an extraneous solution

$\sqrt{10-1} = 10-7$

$\sqrt{9} = 3$

3=3

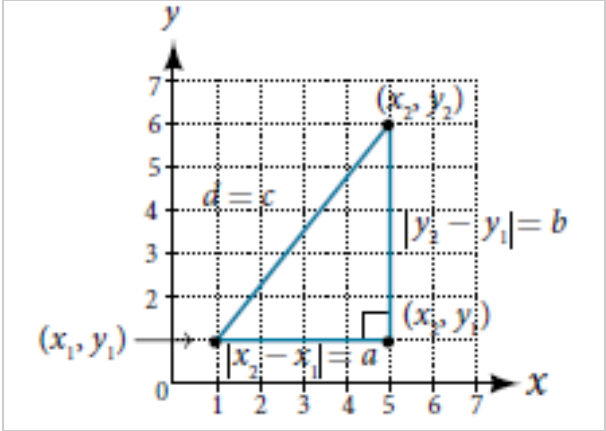
x = 10

The Distance Formula

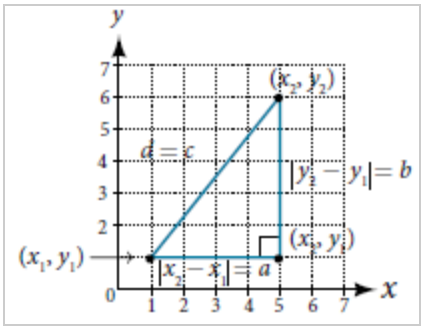
The **distance**, d, between the points (x_1, y_1) and (x_2, y_2) is given by the formula:

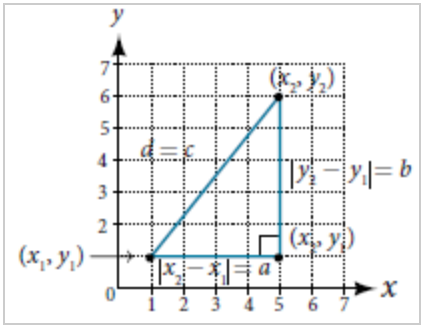
$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

This formula is based directly on the Pythagorean Theorem. When finding the distance between two points, we are actually finding the length of the hypotenuse of a right triangle. The legs of the triangle are $|x_2 - x_1|$ and $|y_2 - y_1|$.

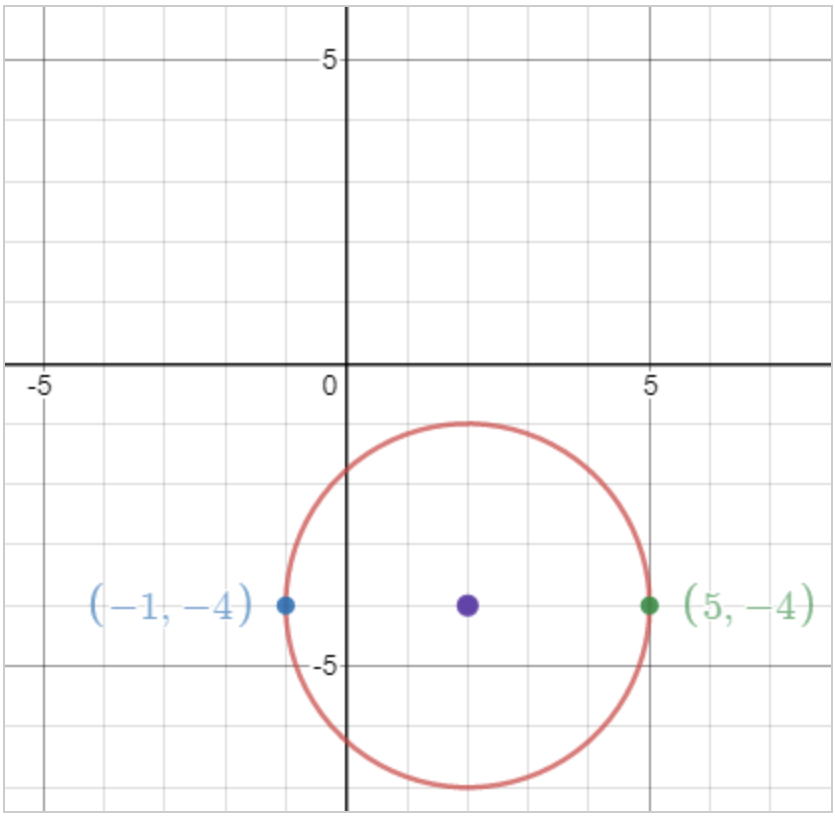


▶ View Description

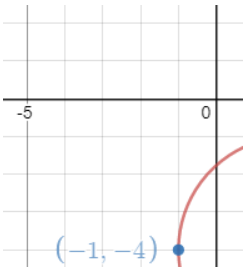




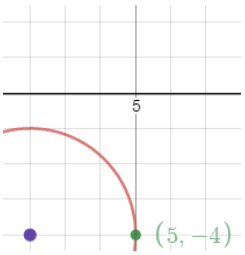
▶ View Description



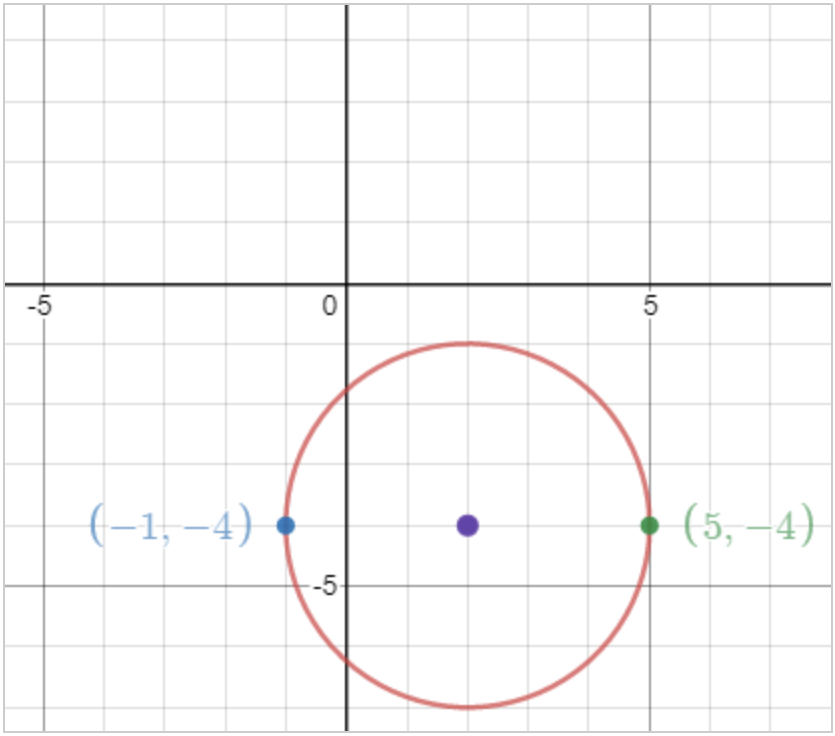
-5



(-1,-4) 5 0 5

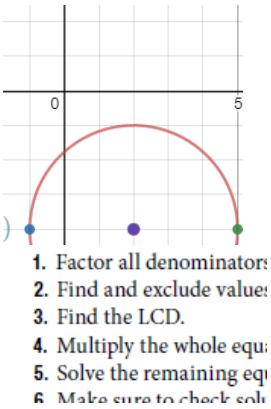


(5,-4) -5



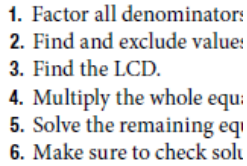
▶ *View Description*

1.



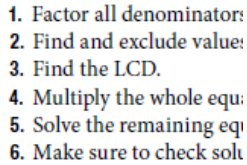
Factor all denominators in the equation.

2.



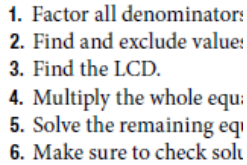
Find and exclude values that set each denominator equal to zero.

3.



Find the LCD.

4.



Multiply the whole equation by the LCD. If the LCD is correct, there will be no denominators left.

5.

1. Factor all denominators
2. Find and exclude values
3. Find the LCD.
4. Multiply the whole equa
5. Solve the remaining equ
6. Make sure to check solu

Solve the remaining equation.

6.

1. Factor all denominators
2. Find and exclude values
3. Find the LCD.
4. Multiply the whole equa
5. Solve the remaining equ
6. Make sure to check solu

Make sure to check solutions back in the original equations to avoid a solution producing zero in a denominator.

34. The slope is $\frac{3}{4}$

he slope is $\frac{3}{4}$ and it passes through the

$-4=\frac{3}{4}(x-1)$ Distribute the 3/4

$-4=\frac{3}{4}x-\frac{3}{4}$ Add 4 to both sides

and it passes through the point (1,4).

$y-4=\frac{3}{4}$

34. The slope is $\frac{3}{4}$ and it passes thr

$y-4=\frac{3}{4}(x-1)$ Distribute the

$y-4=\frac{3}{4}x-\frac{3}{4}$ Add 4 to both

$(x-1)$ Distribute the 3/4

$y-4=\frac{3}{4}$

34. The slope is $\frac{3}{4}$ and it passes t

$y-4=\frac{3}{4}(x-1)$ Distribute tl

$y-4=\frac{3}{4}x-\frac{3}{4}$ Add 4 to bot

3 13

$x-\frac{3}{4}$

4. The slope is $\frac{3}{4}$ and it passes through

$y-4=\frac{3}{4}(x-1)$ Distribute the 3/4

$y-4=\frac{3}{4}x-\frac{3}{4}$ Add 4 to both side

3 13

Add 4 to both sides

$y=\frac{3}{4}$

he slope is $\frac{3}{4}$ and it passes through th

$-4=\frac{3}{4}(x-1)$ Distribute the 3/4

$-4=\frac{3}{4}x-\frac{3}{4}$ Add 4 to both sides

3 13

$x+\frac{13}{4}$

e slope is $\frac{3}{4}$ and it passes through the

$-4=\frac{3}{4}(x-1)$ Distribute the 3/4

$-4=\frac{3}{4}x-\frac{3}{4}$ Add 4 to both sides

$=\frac{3}{4}x+\frac{13}{4}$

54. $125=0.006A^2-0.02A+120$

$0.006A^2-0.02A-5=0$

Subtract 125 from both sides

Use the quadratic formula

$$A=\frac{-(-0.02)\pm\sqrt{(-0.02)^2-4(0.006)(-5)}}{2(0.006)}$$

$=0.006A^2-0.02A+120$	Subtract 125 fr
$06A^2-0.02A-5=0$	Use the quadrat
$\frac{-(-0.02)\pm\sqrt{(-0.02)^2-4(0.006)(-5)}}{2(0.006)}$	Multiply
$\frac{0.02\pm\sqrt{0.0004+0.12}}{0.012}$	Add
$\frac{0.02\pm\sqrt{0.1204}}{0.012}$	Simplify

▶ *View Description*

$.006A^2-0.02A+120$	S
$^2-0.02A-5=0$	U
$\frac{-0.02)\pm\sqrt{(-0.02)^2-4(0.006)(-5)}}{2(0.006)}$	M
$\frac{02\pm\sqrt{0.0004+0.12}}{0.012}$	A
$\frac{02\pm\sqrt{0.1204}}{0.012}$	Si
$06A^2-0.02A+120$	Su
$-0.02A-5=0$	Us
$\frac{0.02)\pm\sqrt{(-0.02)^2-4(0.006)(-5)}}{2(0.006)}$	Mu
$\frac{\pm\sqrt{0.0004+0.12}}{0.012}$	Ad
$\frac{\pm\sqrt{0.1204}}{0.012}$	Sin

Multiply

$$A=\frac{0.02\pm\sqrt{0.0004+0.12}}{0.012}$$

$=0.006A^2-0.02A+120$	Subtract 125
$06A^2-0.02A-5=0$	Use the quac
$\frac{-(-0.02)\pm\sqrt{(-0.02)^2-4(0.006)(-5)}}{2(0.006)}$	Multiply
$\frac{0.02\pm\sqrt{0.0004+0.12}}{0.012}$	Add
$\frac{0.02\pm\sqrt{0.1204}}{0.012}$	Simplify
30.58	

▶ *View Description*

$0.006A^2 - 0.02A + 120$

$6A^2 - 0.02A - 5 = 0$

$\frac{-(-0.02) \pm \sqrt{(-0.02)^2 - 4(0.006)(-5)}}{2(0.006)}$

$\frac{0.02 \pm \sqrt{0.0004 + 0.12}}{0.012}$

$\frac{0.02 \pm \sqrt{0.1204}}{0.012}$

30.58

$0.006A^2 - 0.02A + 120$

$A^2 - 0.02A - 5 = 0$

$\frac{(-0.02) \pm \sqrt{(-0.02)^2 - 4(0.006)(-5)}}{2(0.006)}$

$\frac{.02 \pm \sqrt{0.0004 + 0.12}}{0.012}$

$\frac{.02 \pm \sqrt{0.1204}}{0.012}$

0.58

Add

$$A = \frac{0.02 \pm \sqrt{0.1204}}{0.012}$$

$= 0.006A^2 - 0.02A + 120$	Subtract 1
$06A^2 - 0.02A - 5 = 0$	Use the qu
$\frac{-(-0.02) \pm \sqrt{(-0.02)^2 - 4(0.006)(-5)}}{2(0.006)}$	Multiply
$\frac{0.02 \pm \sqrt{0.0004 + 0.12}}{0.012}$	Add
$\frac{0.02 \pm \sqrt{0.1204}}{0.012}$	Simplify
30.58	
years old	

▶ *View Description*

$0.006A^2 - 0.02A + 120$

$6A^2 - 0.02A - 5 = 0$

$\frac{-(-0.02) \pm \sqrt{(-0.02)^2 - 4(0.006)(-5)}}{2(0.006)}$

$\frac{0.02 \pm \sqrt{0.0004 + 0.12}}{0.012}$

$\frac{0.02 \pm \sqrt{0.1204}}{0.012}$

30.58

ears old

$0.006A^2 - 0.02A + 120$

$A^2 - 0.02A - 5 = 0$

$\frac{(-0.02) \pm \sqrt{(-0.02)^2 - 4(0.006)(-5)}}{2(0.006)}$

$\frac{.02 \pm \sqrt{0.0004 + 0.12}}{0.012}$

$\frac{.02 \pm \sqrt{0.1204}}{0.012}$

0.58

ars old

Simplify

A ≈ 30.58

$0.006A^2 - 0.02A + 120$

$6A^2 - 0.02A - 5 = 0$

$\frac{-(-0.02) \pm \sqrt{(-0.02)^2 - 4(0.006)(-5)}}{2(0.006)}$

$\frac{0.02 \pm \sqrt{0.0004 + 0.12}}{0.012}$

$\frac{0.02 \pm \sqrt{0.1204}}{0.012}$

30.58

ears old

31 years old 24.

24. $\sqrt{t+1}+9=7$
$\sqrt{t+1}+9=7$ Subtract 9 f
$\sqrt{t+1}=-2$ Note that th
No solution

▶ *View Description*

$\sqrt{t+1}+9=7$

24. $\sqrt{t+1}+9=7$
$\sqrt{t+1}+9=7$ Subtract 9 f
$\sqrt{t+1}=-2$ Note that th
No solution

▶ *View Description*

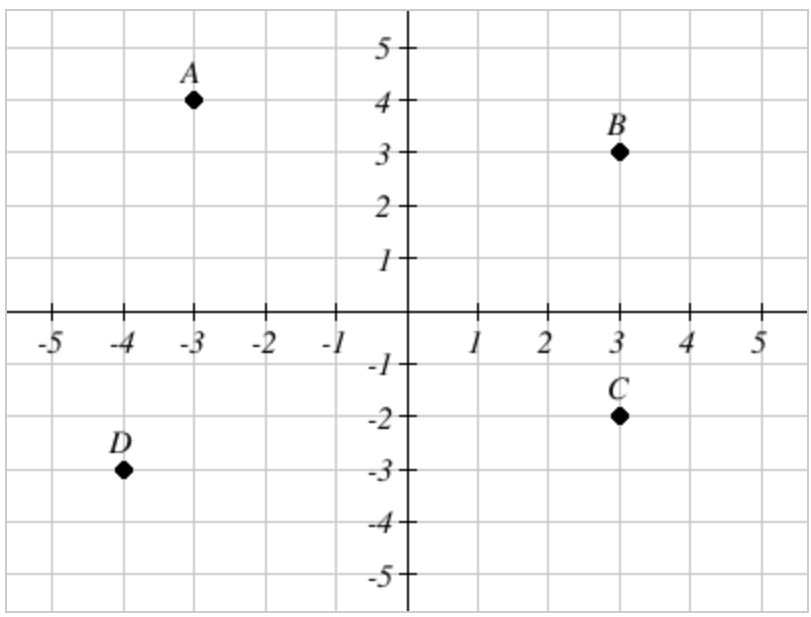
$\sqrt{t+1}+9=7$ Subtract 9 from both sides

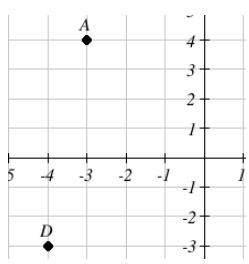
24. $\sqrt{t+1}+9=7$
$\sqrt{t+1}+9=7$ Subtract 9
$\sqrt{t+1}=-2$ Note that t
No solution

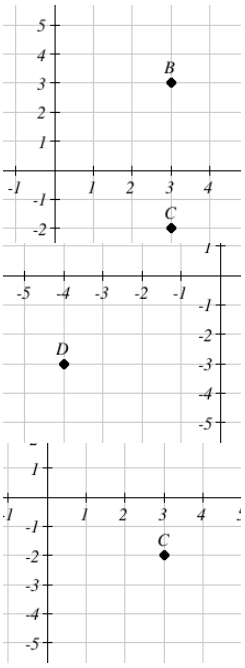
▶ *View Description*

$\sqrt{t+1}=-2$ Note that the square root is equal to a negative value No solution

Points on a Coordinate Plane







22.

22. $\left(\frac{-5+4}{2}, \frac{-6+2}{2}\right)$ Add

$\left(\frac{-1}{2}, \frac{-4}{2}\right)$ Simplify

$\left(\frac{-1}{2}, -2\right)$

▶ View Description

$\left(\frac{-5+4}{2}, \frac{-6+2}{2}\right)$ Add

22. $\left(\frac{-5+4}{2}, \frac{-6+2}{2}\right)$ Add

$\left(\frac{-1}{2}, \frac{-4}{2}\right)$ Simplify

$\left(\frac{-1}{2}, -2\right)$

▶ View Description

$\left(\frac{-1}{2}, \frac{-4}{2}\right)$ Simplify

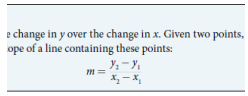
22. $\left(\frac{-5+4}{2}, \frac{-6+2}{2}\right)$ Add

$\left(\frac{-1}{2}, \frac{-4}{2}\right)$ Simplify

$\left(\frac{-1}{2}, -2\right)$

▶ View Description

$\left(\frac{-1}{2}, -2\right)$ the slope of a line The slope of a line, m, represents the change in y over the change in x. Given two points, (x_1, y_1) and (x_2, y_2), the following formula determines the slope of a line containing these points:

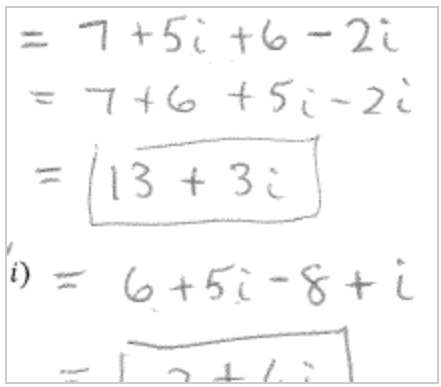


▶ View Description

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

= 7+5*i*+6-2*i*

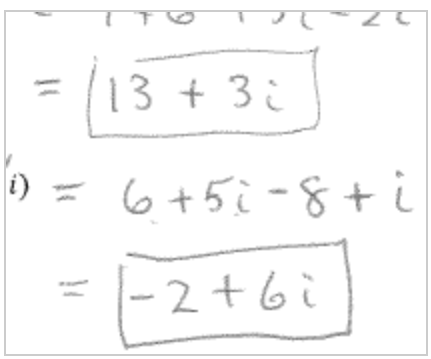
= 7+6 +5*i*-2*i*



▶ View Description

= 13+3*i*

= 6+5*i*-8+*i*



▶ View Description

= -2+6*i*

9. $(x-1)^{\frac{3}{4}} = 8$

$(x-1)^{\frac{3}{4}} = 8$ Rewrite

9. $(x-1)^{\frac{3}{4}}=8$

$(x-1)^{\frac{3}{4}}=8$	Rewrite
$\sqrt[4]{(x-1)^3}=8$	Raise bot
$\left(\sqrt[4]{(x-1)^3}\right)^4=8^4$	Take the
$\sqrt[3]{(x-1)^3}=\sqrt[3]{8^4}$	Evaluate
$x-1=2^4$	Evaluate

$\sqrt[4]{(x-1)^3}=8$ Raise both sides to the 4th power

9. $(x-1)^{\frac{3}{4}}=8$

$(x-1)^{\frac{3}{4}}=8$	Rewrite
$\sqrt[4]{(x-1)^3}=8$	Raise both
$\left(\sqrt[4]{(x-1)^3}\right)^4=8^4$	Take the c
$\sqrt[3]{(x-1)^3}=\sqrt[3]{8^4}$	Evaluate tl
$x-1=2^4$	Evaluate
$x-1=16$	Add 1 to b

$(\sqrt[3]{(x-1)})^4=8^4$ Take the cube root of both sides

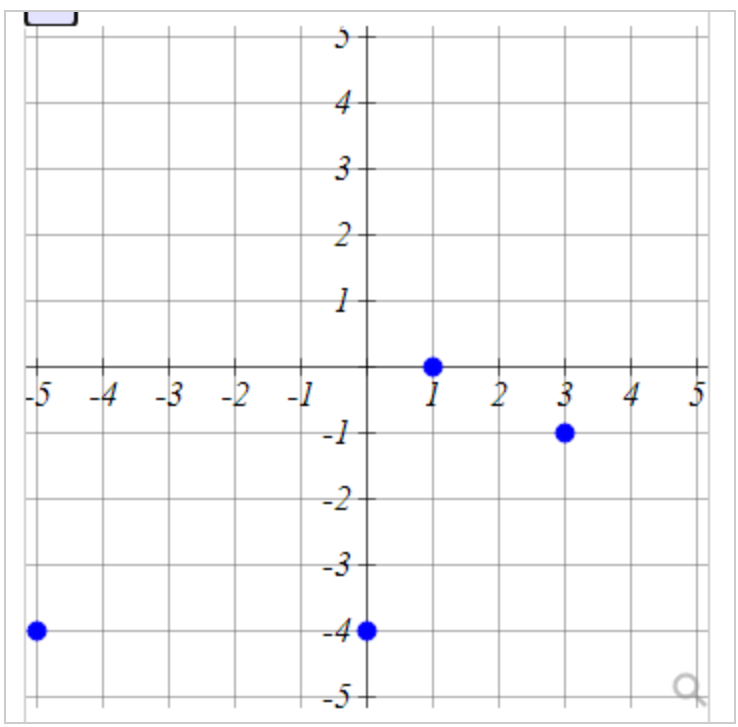
$(x-1)^{\frac{3}{4}}=8$	Rewri
$\sqrt[4]{(x-1)^3}=8$	Raise t
$\left(\sqrt[4]{(x-1)^3}\right)^4=8^4$	Take t
$\sqrt[3]{(x-1)^3}=\sqrt[3]{8^4}$	Evalua
$x-1=2^4$	Evalua
$x-1=16$	Add 1 t
$x=17$	

$\sqrt[3]{(x-1)}=\sqrt[3]{8^4}$ Evaluate the cube root

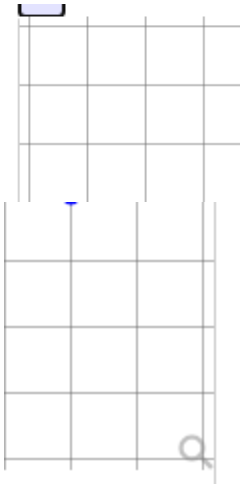
$x-1=2^4$ Evaluate

$x-1=16$ Add 1 to both sides

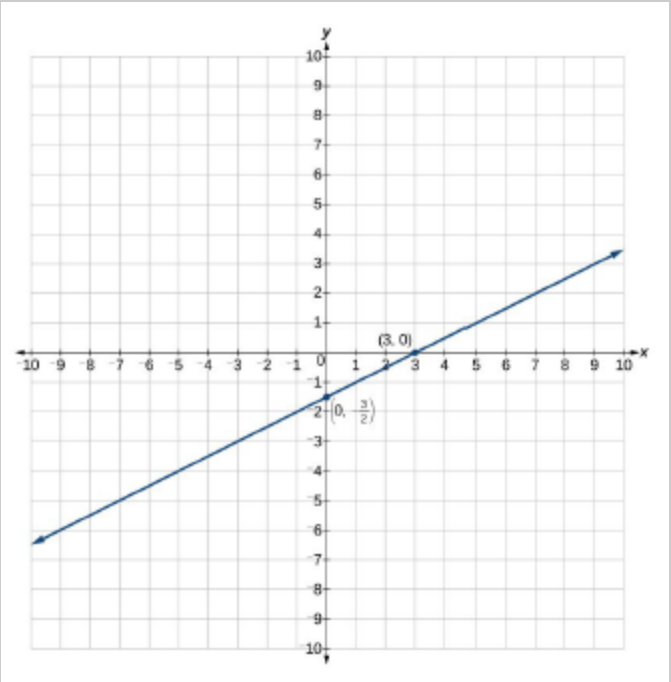
$x=17$



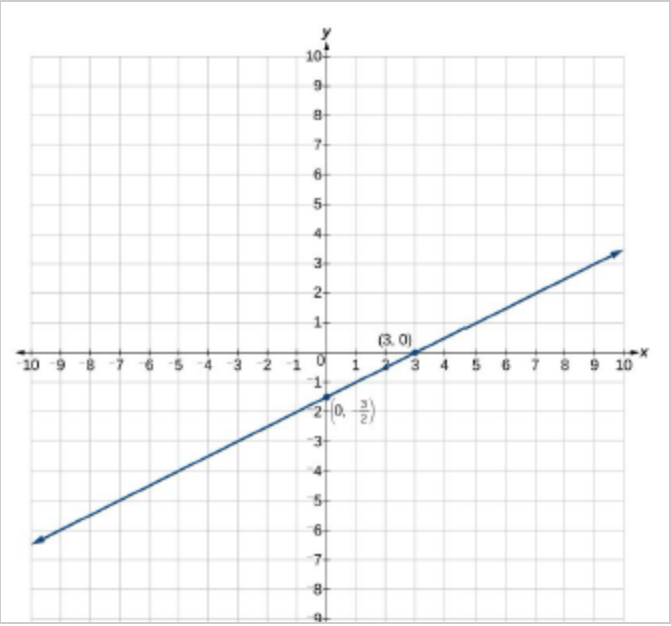
-5 -4 -3 -2 -1
5
4
3
2
1
1
2 3
4
5
-1
-2
-3
-4
-5



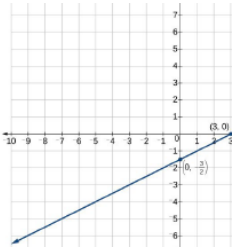
Linear Functions

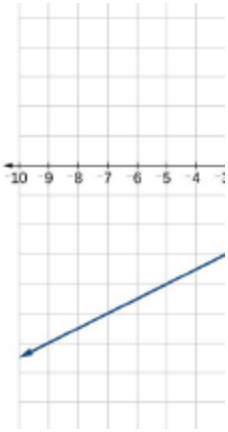


[Note: Graph of a linear function]

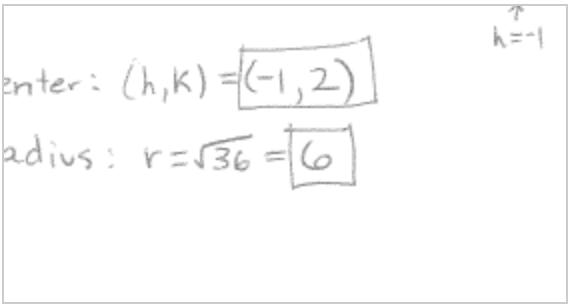


▶ *View Description*



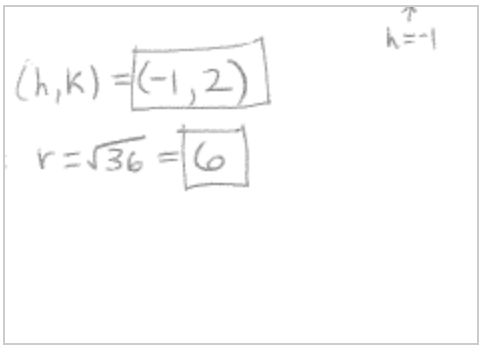


Center: $(h, k) = \overline{(-1,2)}$



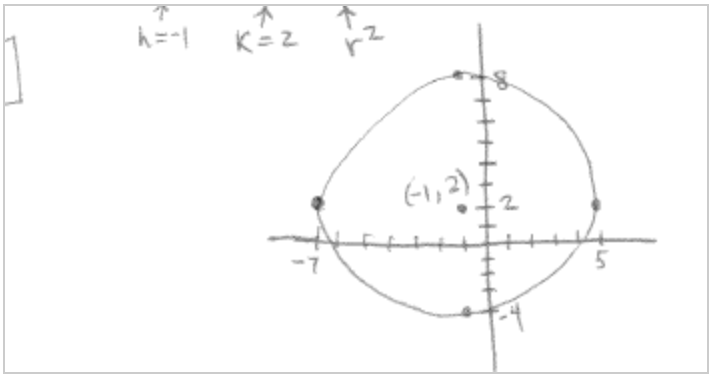
▶ *View Description*

radius: $r = \sqrt{36} = \overline{6}$



▶ *View Description*

h=-1 K=2 r^2



▶ *View Description*

(-1,2) 2 5

> ***rational equations***
> A **rational equation** contains at least one rational expression where the variable appears in at least one of the denominators.

rational equations

A rational equation contains at least one rational expression where the variable appears in at least one of the denominators. 19. $\frac{3x}{x-1}+2=\frac{3}{x-1}$

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$\frac{3x}{x-1}+2=\frac{3}{x-1}$ Mu

$\frac{3x}{x-1}\cdot x-1+2(x-1)=$

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$\frac{3x}{x-1}+2=\frac{3}{x-1}$ Multiply both sides by x-1

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19. $\frac{3x}{x-1}+2=\frac{3}{x-1}$

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$3x+2x-2=3$ Con

$=\frac{3}{x-1}$

$=\frac{3}{x-1}$ Multiply both sides by x-1

$-1+2(x-1)=\frac{3}{x-1}\cdot x-1$ Simplify

$-2=3$ Combine like terms

$= \frac{3}{x-1}$

$= \frac{3}{x-1}$ Multiply both sides by x-1

$-1+2(x-1) = \frac{3}{x-1} \cdot x-1$ Simplify

$-2 = 3$ Combine like terms

$\frac{3x}{x-1} \cdot x - 1 + 2(x-1) = \frac{3}{x-1} \cdot x - 1$ Simplify

3x+2x-2=3 Combine like terms

5x-2=3 Add 2 to both sides

5x=5 Divide both sides by 5

x≠1; when we solve this we get x = 1, which is excluded, therefore NO solution the **point-slope formula** Given one point and the slope, the point-slope formula will lead to the equation of a line:

the point-slope formula

Given one point and the slope, the point-slope formula will lead to the equation of a line

$$y - y_1 = m(x - x_1)$$

▶ *View Description*

$$y - y_1 = m(x - x_1)$$

$=$

$= \boxed{5 - i\sqrt{15}}$ (OR

Real: 5

Imag: $-\sqrt{15}$

▶ *View Description*

$5 - i\sqrt{15}$ (OR $5 + (\sqrt{15})i$) (OR $5 + (-\sqrt{15})i$)

Real: 5

Imag: $-\sqrt{15}$

$i\sqrt{32} = i\sqrt{16}\sqrt{2} =$

(OR $0 + (-\sqrt{15})i$)

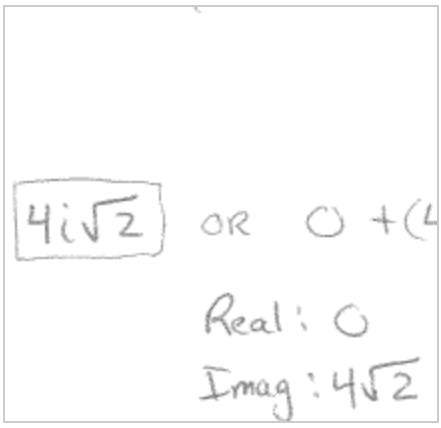
) OR $0 + (4\sqrt{2})i$

Real: 0

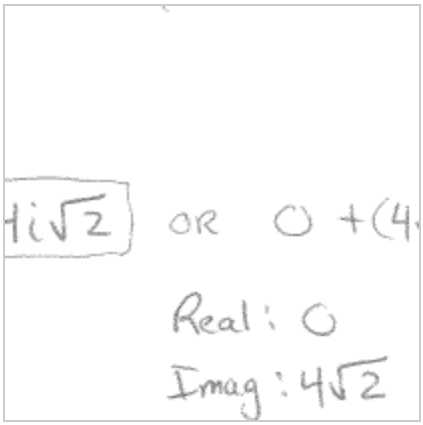
Imag: $4\sqrt{2}$

▶ View Description

$4i\sqrt{2}$ OR $0 + (4\sqrt{2})i$



▶ View Description



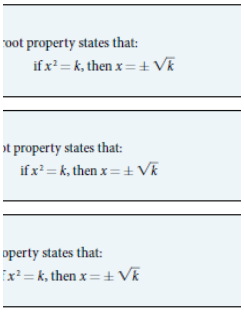
▶ View Description

Real: 0

Imag: $4\sqrt{2}$

the square root property

With the x² term isolated, the square root property states that:



if $x^2 = k$, then $x = \pm\sqrt{k}$ where *k* is a nonzero real number.

1. Isolate the radical expression on one side of the equal sign. Put all remaining terms on the other side.
2. If the radical is a square root, then square both sides of the equation. If it is a cube root, then raise both sides of the equation to the third power. In other words, for an nth root radical, raise

both sides to the nth power. Doing so eliminates the radical symbol.
3. Solve the remaining equation.
4. If a radical term still remains, repeat steps 1-2.
5. Confirm solutions by substituting them into the original equation.

▶ *View Description*

▶ *View Description*

▶ *View Description*

y-axis Quadrant II Quadrant I x-axis Quadrant III Quadrant IV The xy-plane and its four quadrants A labeled coordinate system Plotting the ordered pair (3,-1)

▶ *View Description*

▶ *View Description*

6.

$x^{\frac{2}{3}} = 16$
$x^{\frac{2}{3}} = 16$ Rewrite

$\sqrt[3]{x^2} = 16$ Cube both sides

$(\sqrt[3]{x^2})^3 = 16^3$ Evaluate
$x^2 = 16^3$ Take the square root of both sides

$\sqrt{x^2} = \sqrt{16^3}$

Use the square root property
$x = \pm 4^3$ Evaluate
$x = \pm 64$ given an equation, find the intercepts.

- Find the x-intercept by setting $y = 0$ and solving for x.
- Find the y-intercept by setting $x = 0$ and solving for y.

18. $d = \sqrt{(7-2)^2 + (4-(-5))^2}$ Subtract

▶ View Description

$d = \sqrt{(5)^2 + (9)^2}$ Square the 5 and 9

▶ View Description

$d = \sqrt{25 + 81}$ Add

▶ View Description

$d = \sqrt{106}$

▶ View Description

6. $7x + 2 = 3x - 9$
$7x + 2 = 3x - 9$ Subtract 3x from both sides
$4x + 2 = -9$ Subtract 2 from both sides
$4x = -11$ Divide both sides by 4
$x = \frac{-11}{4}$

▶ View Description

Graph

▶ View Description

$y\ y = -3x - 2\ y = 2x + 1$ -12-10 68 $y = \frac{1}{3}x + 2\ x$

Slope Intercept Form

Slope-Intercept Form: $y = mx + b$

- m = Slope
- b = y-intercept

Example: $y = 3$

▶ View Description

[Note: y = 0x + 3 m = 0 (no slope)]

a) $\frac{(2-i)}{(2+5i)} \cdot \frac{(2-5i)}{(2-5i)} = \frac{4-10i-2i+5i^2}{4-10i+10i-25i^2} = \frac{4-12i+5(-1)}{4-25(-1)} = \frac{-1-12i}{29}$

Complex
Conjugate: 2-5i

OR

$\frac{-1}{29} - \frac{12}{29}i$ [GRAPH_BBOX: 101, 83, 109, 914, graph, A number line segment, with an open circle at 1, a closed circle at 5, and a blue line connecting them.]

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